

IllumiWand: Improving 3D Interaction with Monoscopic Displays Through a Projected Physical 3D Pointer

Philipp Löschner*

André Hinkenjann†

Ernst Kruijff‡

Institute of Visual Computing
Bonn-Rhein-Sieg University of Applied Sciences
Sankt Augustin, Germany

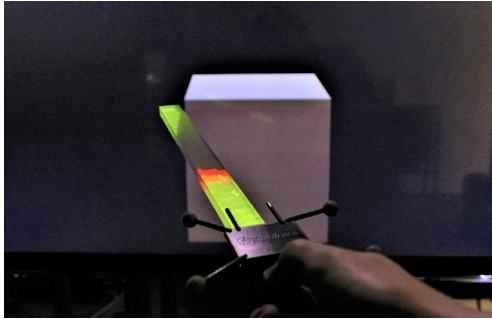


Figure 1: IllumiWand prototype intersecting a virtual cube (green: visible part of rod, red: collision, transparent: non visible part of rod).

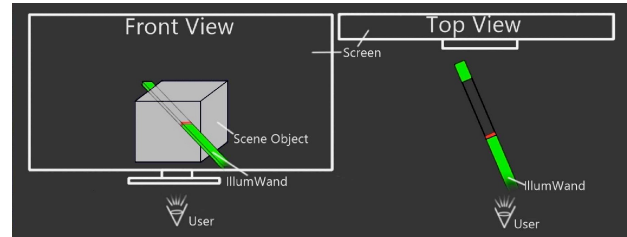


Figure 2: Schematic view of the spatial situation of the IllumiWand intersecting a cube shown on the display (cf. fig. 1).

ABSTRACT

This work presents a system concept and implementation that targets the improvement of spatial perception when interacting with virtual objects displayed perspectively on a monoscopic display. The IllumiWand is an illuminated rod that can be partially made transparent to occlude or show (parts of) virtual objects the rod intersects with. As the rod is operated in front of the 2D display, it affords spatial perception of objects in the volume that can be swept with it between the user and the display. Our concept extends previous work on hover space systems by using the volume in front of a display as an additional medium to provide spatial visual cues.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Pointing devices

1 INTRODUCTION

Displays are often restricted to monoscopic output even when interacting with 3D content, caused for example by cost considerations or technical limitations. When interacting with 3D scenes on a 2D display, a virtual laser beam (ray casting) is typically used to select and manipulate objects [2]. While selecting objects is reasonably straightforward, comprehending the spatial properties of the shown objects can be quite difficult when using monoscopic displays. In this paper, we present a first implementation and study of a hybrid 2D/3D system that deploys 3D interaction with monoscopically presented but perspectively rendered interactive content using a projected – hence illuminated – rod called IllumiWand (cf. fig. 1). The

concept behind IllumiWand is showing the intersection of a pointing device – in particular the ray extending from the pointer – with scene objects in a spatially correct manner, in the 3D volume in front of the 2D display. This can principally be achieved by a physical device that can show the ray cast by the pointer while paying attention to occlusions of the ray with scene objects, in dependence to the users perspective. To this extent, IllumiWand is a physical pointer representing a color-variant fixed-length ray – the virtual laser beam representing a cast pick ray. It consists of a semi-transparent rod which is projected upon using a top-down projection system. As such, it visualizes the interaction of the ray with objects in the scene. If the ray intersects with a scene object – as illustrated in fig. 1, 2 – it shows the following: (a) the parts of the object occluded (intersected) by the ray as can be observed through the transparent rod, and (b) parts of the ray that are or not occluded by the object, visualized through color coding in green (parts of the ray not intersecting with the object) and red (the front intersection between object and ray).

If the ray does not intersect with any object, it is completely green. As such the user is able to interact with the perspectively presented object at its position in real space: the IllumiWand extends the 2D object displayed on the display with spatial cues on the rod itself. Doing so, the IllumiWand presents additional spatial information in front of the 2D display: by sweeping the rod through space, spatial (perspectively correct) cues associated with shown objects can be perceived throughout the volume in front of the 2D display, albeit only in the section of space occupied by the ray. As a result, the spatial properties of the object in 3D space displayed in 2D can principally be deduced. Hence, we assume this interaction approach leads to better understanding of the spatial properties (e.g., size and distance) of the scene's objects. We explore this in a pilot study reported in this paper. Some properties of our approach may remind of hover space approaches that make use of the space over in particular table-top displays [3]. Most of the hover space approaches have looked at affording additional functionality (e.g. menu systems), or providing haptic feedback. To our knowledge, the space above or in front of the display has not been used in a way similar to our system, to add spatial visual cues to objects shown on a 2D display, as most approaches have rather focused on other

*Philipp.Loeschner@h-brs.de

†Andre.Hinkenjann@h-brs.de

‡Ernst.Kruijff@h-brs.de

properties, e.g., haptics [4].

2 DESIGN SPACE AND IMPLEMENTATION

There are several possibilities to realize the IllumiWand concept as a physical system. We considered two of them in more detail.

Self illuminated device In this variant the pointing device is a hand-held active display with the form factor of a wide, truncated laser beam. Such a display could show the beam as well as parts of objects. It is potentially very bright and offers high contrast. On the other hand there are some disadvantages: It is difficult to find a display that is long and very small and it is difficult to calibrate the brightness and color in order to match the content on the screen. Also, the display has to be driven by extensive electronics which need to be part of the pointing device. Considering the advantages and disadvantages, we concluded that this realization is non-optimal.

Passive pointer Here, the pointer is a rigid transparent canvas that is projected on (cf. fig.1). The laser beam is shown by projecting on the parts of the physical device that are not occluded by objects in virtual space. To display parts of objects that are occluding the pointer, nothing is projected on the device, leaving the transparent canvas to show the objects presented perspectively correct on the screen (objects shine through). By this, all active parts are omitted from the pointer. Color and brightness of the scene objects are always correct, since only the display color and brightness are seen for scene objects.



Figure 3: System setup with short throw projector (top middle), camera based tracking-system (top), monoscopic display with virtual cube displayed (middle) and the IllumiWand realized as a tracked handle with an acrylic rod covered with holographic foil (right)

Considering the above, we decided upon the following implementation. The system consists of several main components (cf. fig. 3): a transparent pointing device (rod) to which a tracking target is attached, a tracking setup (OptiTrack) to track the interaction device and the user's head (using a bike helmet and a tracking target), a short-throw projector (2800 ANSI lumens) to project down on the pointing device, and a 47 inch 2D display. The pointing device consists of an acrylic glass rod of 50 cm length (including a handle of length 14 cm) and 2,5 cm width and 1 cm height, to which a transparent projection foil is attached to all sides. Projection mapping [1] is used to achieve a visible color-change on the interaction device. This method is used to keep the interaction device as small

and simple as possible. The projector and the tracking system are mounted on the top level of an aluminium structure. The viewing angle of the tracking cameras has been set as such that both the interaction device and the user can be tracked simultaneously.

Unity Version 2018.1 is used to render the 3D environment and the projection mapping to render (project) colors on the interaction device. The Unity scene includes 3 cameras. One is looking at the scene from the viewpoint of the projector (used for the projection mapping), one from the user perspective (shown on the LCD). A third camera is used later as additional screen for the pilot study. The tracking of the user is positional only, no rotational values are used because the view plane (i.e. the display) is fixed in space.

For testing purposes, the user interacts with a virtual cube positioned in the volume in front of the 2D display, hence between the display and the user. The virtual interaction device is segmented into 71 x 5 segments (5 with and 71 length). Segmentation is used to check for occlusion and collision with the objects in the scene. If none are detected, the segments are green. To be able to perceive the rod/object intersection better, we added a small red highlight at the spot where the rod intersects with the object for the first time (cf. fig. 1, 2). If a collision is detected the segments are red and when occluded the segments are not projected resulting in transparency at that spot. Alternatively, when removing the acrylic glass rod from the handle and deactivate projection mapping it can be used as a pure virtual pointer (cf. fig. 4), showing a virtual laser beam emerging from the handle on the screen instead of on the rod.

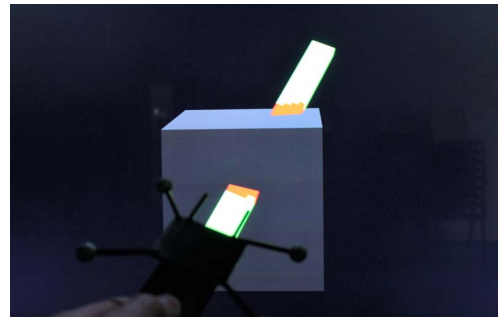


Figure 4: Interaction device with virtual pointer. Only the tracked handle is used, the rod is virtual (part of the displayed image.)

3 PILOT STUDY

To create a better understanding of the performance of the system, we performed a first pilot study. In particular, we looked at whether the projected interaction device could improve spatial judgments in relation to the virtual object the user interacted with, by focusing on its size and position. We expected the IllumiWand to be better suited for this tasks than the pure virtual pointer, leading to better results in comparison to a pointer that could only rely on visual cues displayed on the monoscopic display. Based on the results of the pilot study, further studies are planned.

3.1 Design and procedure

The pilot study uses all the components described in section 2. Test scenes contained a cube as basic geometric object for interactions and spatial judgments. In every scene the cube would appear at different locations and in different sizes. Users judged the position and size while using two different interaction modes: the rod with additional projection (IllumiWand) and a virtual pointer with visual cues solely displayed on the monoscopic display. We logged selection

time and position and observed the users during the performance of the task, while making notes.

Each test involved two tasks: estimating the location and size of the object, to analyse potential improvements in spatial judgments. Prior to starting the tests each test person was familiarized with the functionality and the behavior of the system. The cubes in the test scenes should be placed inside the test persons interaction space and at the height of the LCD display. The study consisted of 10 trials with each interaction device. Users performed the task first with the IllumiWand, followed by the virtual pointing device. The order of the test scenes was randomized. Users started from a fixed spot marked on the ground. The first part of the test was to judge the position of the front face of the object. The second part of the test was to estimate the size of the cube. Two wooden rods (x, y, z axis) with sectors from 1 to 12 (1 sector = 10cm) were used to help the participants for estimating size and position (cf. fig. 3). Both tests were timed and users were informed to perform the task as quickly but also as precisely as possible. In both cases the test person vocally noted when they were sure about finding the front or the size. After all 20 repetitions the participant was asked to give subjective feedback through a survey, consisting of questions about usability, long-term usage, posture and preferred variant interaction device. The questionnaire used a Likert scale, from 1 (negative) - 7 (positive).

3.2 Results

20 people participated in the study. The age of the participants ranged from 22 to 40 years (4 female). All participants except one had a computer science background. A T-test for dependent samples was used to compare results of both conditions. It turned out that users could estimate the object size more precisely using the IllumiWand ($M = 2.1, SD = 1.44$) than with the virtual pointer ($M = 5.2, SD = 2.56$), $t(19) = -5.15, p < .001$ (cf. fig. 5).

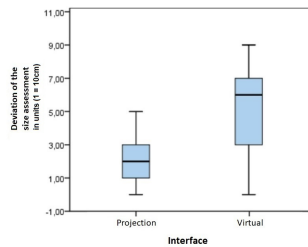


Figure 5: Performance of the size estimate

The location of the virtual object was assessed faster with the virtual pointer ($M = 5.34, SD = 1.34$) than with the IllumiWand ($M = 7.87, SD = 1.60$), $t(19) = 10.7, p < .001$ (cf. fig. 6).

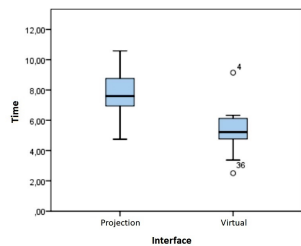


Figure 6: Time needed to find the front of the object

The position of the object was estimated correctly every time by

the user regardless of the used method. Only when using the virtual pointer did the performance differ significantly for size estimation between repetitions. The projection condition shows an effect between repetition and the required time. In the first repetition, the users needed significantly longer than in all subsequent repetitions, showing potential learning effects. When using the virtual pointer, the time required differs between different repetitions, but without a trend becoming visible.

Subjective data was obtained using an online questionnaire. The usability of the virtual pointer ($M = 5.7$) was evaluated better by the test users than the projection ($M = 4.4$). However, for long-term use the virtual pointer ($M = 6$) is rated slightly better than the projection ($M = 5.75$), likely because it is lighter in weight. The preferred variant of most test users is the virtual pointer: ($M = 5.6$) for virtual pointer, ($M = 4.85$) for projection. The reviews for the assessment of the position do not differ significantly between the projection ($M = 5.05, SD = 0.95$) and the virtual pointer ($M = 4.95, SD = 1.15$). However, when assessing the size, users have the impression that the projection is better suited for this (cf. fig. 5 ($M = 5.65, SD = 1.14$), than the virtual pointer ($M = 4.70, SD = 1.08$), $t(19) = 3.13, p = .005$). The projection was also rated more helpful for object perception ($M = 6.1, SD = 1.55$) than the virtual rod ($M = 5.35, SD = 1.27$), $t(19) = 2.6, p = 0.18$). It was stated that it allows for a better assessment of the spatial awareness of the virtual object ($M = 6.1, SD = 1.71$ vs. $M = 4.55, SD = 1.32$), $t(19) = 4.84, p < .001$).

3.3 Discussion

Results of the pilot study indicate that most users needed more time to complete the tasks with the projection than with the virtual pointer. This result is reminiscent of a finding by Teather Stuerzlinger (2014), where additional depth information improved accuracy at the cost of speed [6]. Yet, the data also shows that users had a better spatial understanding of the virtual objects when using the projection. As such, we show that our technical approach principally can aid 3D interaction with objects shown on a 2D display by providing spatial cues over the IllumiWand. While for positioning, the virtual pointer also worked reasonably well, size judgments were much better with the IllumiWand. We assume this is in part due to the fact that spatial properties do not need to be deduced from 2D content while exploring properties of the objects at the display. In particular, we observed users exploring the edges of an object using IllumiWand to obtain a spatial understanding of the object properties in space, hence, to understand its actual scale.

Test subjects lean more to the virtual variant in terms of finding the position of an object. In terms of general use, subjects were neutral towards both versions, but not totally convinced by either of them. The evaluation showed that most subjects consider the handling of both methods uncomfortable. Usability-wise both variants were rated similarly, but the virtual variant was rated slightly better compared to the projection, likely because of the lower weight caused by the absence of the acrylic rod. Due to the tracking movements like turning the interaction device upside down, rotating it more than 90° or certain tilt movements are not possible, which initially seems unnatural for test subjects. Presumably, prolonged use and the adjustment would minimize this problem. Like the objective data, the subjective feedback regarding object perception of size and spacial awareness was more in favor of the projection. The projection on the interaction device gives the user a good visual feedback regarding the object's shape and size, as well as the possibility of using the projection to scan the object.

4 CONCLUSION AND OUTLOOK

Our pilot study indicates that the IllumiWand interaction device can lead to a better spatial understanding of virtual objects. However, it also showed that the system lacks in certain areas and improvements

would be beneficial. We expected that a comparison of the two pointers – the IllumiWand and the virtual pointer – would show an improvement in spatial perception when using the projection-based solution. On the one hand, the study indeed showed that the system and interaction device with the projection improve the spatial perception of object size and distance. On the other hand, the study showed that many of the tasks could be completed more quickly with the virtual pointer and that the test subjects coped better with the virtual pointer. For example during the pilot study it became evident that the image shown on the display in combination with the IllumiWand was affected by the users dominant eye. Depending on the users the image was slightly off the the left or to the right.

We did not assess the effect of motion parallax while judging the spatial properties, an aspect we intend to cover in a follow-up study. In direct relation, the use of eye tracking could improve the perspectively rendered monoscopic representation of object shown on the LCD screen. The helmet currently used to track the user does not take eye movement into account.

It would also be interesting to also address other types of display technologies, e.g., the usage of foldable display foils around a rod, for example to render different types and higher-resolution feedback and minimize technical overhead by getting rid of the overhead projector. Even more so, we assume that additional non-visual cues, such as vibration, could also improve interaction. Providing a collision cue (initial touch upon an object) could be easily implemented by embedding a vibrotactor in the rod. With higher-density 3D spaces, here it could also be of interest to consider audio-tactile methods to inform about objects that are close by, but not intersecting with the rod, somewhat similar to other approaches shown with vibrotactile gloves [5].

Furthermore, it will be of interest to extend the interface with 3D object manipulation functionality. While a simple on-screen widget could support this, alternative approaches like a widget positioned on top of the IllumiWand (i.e. like a touch-bar) could also be very interesting. Finally, the projection on the IllumiWand pointer could also be used to present user interface elements. Together with finger tracking this could be used to implement a dynamic user interface that is possibly object sensitive (depending on the intersected object).

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